Subliminal Hangover V4

Stage B — Preregistered Mixed-Effects Replication

1 Design

N=594 (**593** pass gate, 1 fail: XTTS speaking_rate below threshold). 3 models (Chatterbox, Kokoro, XTTS-v2) × 10 emotional targets × 10 repetitions. Two conditions: length-matched **noun** primes vs **number** (robotic) primes. Preregistered protocol: `PROTOCOL.md`.

1.1 Per-Condition N

Model × Condition	N
Chatterbox noun	99
Chatterbox number	97
Kokoro noun	100
Kokoro number	100
XTTS noun	100
XTTS number	97

1.2 Mixed-Effects Model

For each model, two random-intercept models:

- `f0_cv ~ condition_num + (1 | target_id) + (1 | repetition)`
- `speaking_rate ~ condition_num + (1 | target_id) + (1 | repetition)`

`condition_num = 0` (noun), `1` (number). Fitted via `statsmodels.MixedLM` with LBFGS optimizer, max 200 iterations.

2 Alignment Quality

WhisperX V3 word alignment: **594** aligned successfully, **12** errors. Aligned segments used for syllable-count-based speaking rate. Target sentence matched via extracted text; f0 extracted via parselmouth.

3 Primary Finding: Tempo Acceleration

`s peaking_rate ~ condition_num` (mixed effects, random intercepts for target_id + repetition)

Model	Coef	95% CI	p	CI check
Chatterbox	+0.352	[+0.103, +0.601]	0.0056**	✓ zero excluded
Kokoro	+1.356	[+1.157, +1.555]	1.5e-40***	✓ zero excluded
XTTS-v2	+0.704	[+0.391, +1.016]	1e-05***	✓ zero excluded

Tempo acceleration confirmed across all three architectures. All 95% CIs exclude zero. The robotic number prime causes a consistent, statistically significant increase in speaking rate ranging from +0.35 (Chatterbox) to +1.36 (Kokoro) syllables/second.

4 Secondary Finding: Pitch Compression

`f0_cv ~ condition_num` (mixed effects, random intercepts for target_id + repetition)

Model	Coef	95% CI	p	Verdict
Chatterbox	-0.006	[-0.031, +0.020]	0.6678ns	Null
Kokoro	-0.045	[-0.061, -0.029]	2.1e-08***	Confirmed
XTTS-v2	-0.004	[-0.030, +0.022]	0.7617ns	Null

Pitch compression is a Kokoro-specific phenotype. Kokoro: coef=-0.045, 95%CI=[-0.061, -0.029], p=2.1e-08 — confirmed. Chatterbox and XTTS show no reliable f0_cv change. Chatterbox f0_cv model has convergence failures; treat as null.

5 Conclusions

1. **Tempo acceleration is robust.** Robotic number primes cause consistent, statistically significant speaking-rate increases across all three tested TTS architectures. Effect sizes range from +0.35 (Chatterbox) to +1.36 (Kokoro) syllables/second, all with 95% CIs excluding zero.

2. **Pitch compression is Kokoro-specific.** Only Kokoro shows a reliable f0_cv drop after robotic primes (coef=-0.045, p=2.1e-08). Chatterbox and XTTS show no reliable pitch compression in this larger-N design.
3. **Two distinct acoustic phenotypes confirmed.** The V3 pilot identified two patterns: pitch compression (Chatterbox, XTTS) and tempo acceleration (Kokoro). V4 finds tempo acceleration in all models and restricts pitch compression to Kokoro — the phenotype map has shifted with increased statistical power.

> **Practical implication:** TTS pipelines should not assume context-independent prosody. Robotic text (numbers, codes, identifiers) bleeds tempo acceleration into adjacent emotional speech. For emotionally-sensitive applications, generate emotionally-targeted segments in isolation rather than as continuations of robotic primes.

6 Limitations

- **Chatterbox f0_cv convergence.** The mixed-effects model for Chatterbox f0_cv failed to converge. Coefficient estimate and p-value are unreliable. Future work should try alternative optimizers (Nelder-Mead, Powell) or Bayesian priors.
- **XTTS seed coupling.** Seed coupling check passed (0/100 identical noun/number seeds), but XTTS internal sentence splitting may still reduce context window bleed vs models that process the full prime+target string jointly.
- **Causal mechanism unclear.** Tempo acceleration could reflect genuine acoustic inertia, attention-like context effects, or token-length artifacts (numbers are shorter tokens than nouns). Controlled token-length analysis is a natural follow-up.
- **Single emotional target style.** All targets are angry/indignant sentences. Whether the hangover generalizes to other emotional styles (sad, happy, neutral) is unknown.
- **Descriptive, not causal.** These are observational effects within TTS context windows. A causal model of how priming affects acoustic production requires controlled parameter manipulation.